



Software Engineering Lab – Flowchart



In Lab 1 **Requirements Analysis**

- IEEE 830 SRS structure
- UC 840 use case structure
- Functional vs Non Functional Requirements
- Problem statement examples



In Lab 2 **UML Use Case Diagrams**

- Actors, system boundary
- Use case notation (<>, <>)
- Associations
- Example diagram (ATM)



In Lab 3 **UML Class Diagrams**

- Class vs domain classes
- Attributes, operations
- Relationships: association, aggregation, composition
- Multiplicity (1..*, 0..1, etc.)



In Lab 4 **Sequence / State / Activity Diagrams**

- Sequence lifelines, messages, state transitions
- State: states, triggers, events
- Activity: swimlanes, decisions, nodes



In Lab 5 **Introduction to Git**

- Git basics (init, add, commit, push, pull)
- Branching, pull request, merge conflict, revert



In Lab 6 **Test-Driven Development (TDD)**

- Write failing test (red)
- Write minimal code (green)
- Refactor
- Project



In Lab 7

- Final Review
- Check